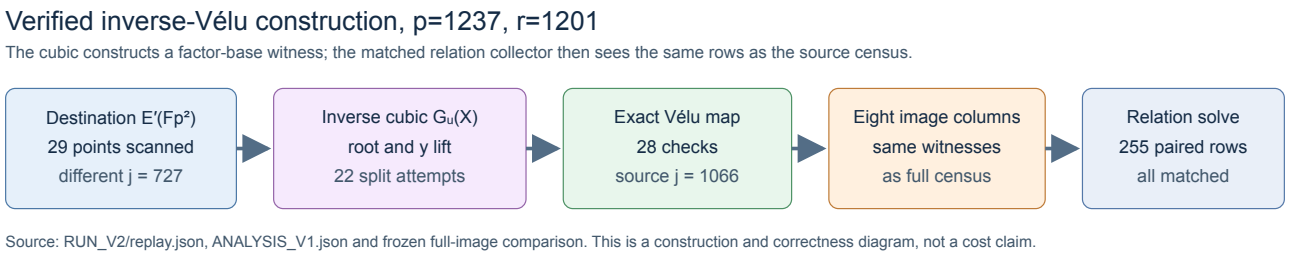
Direct inverse-Vélu fiber oracle: frozen V2 run

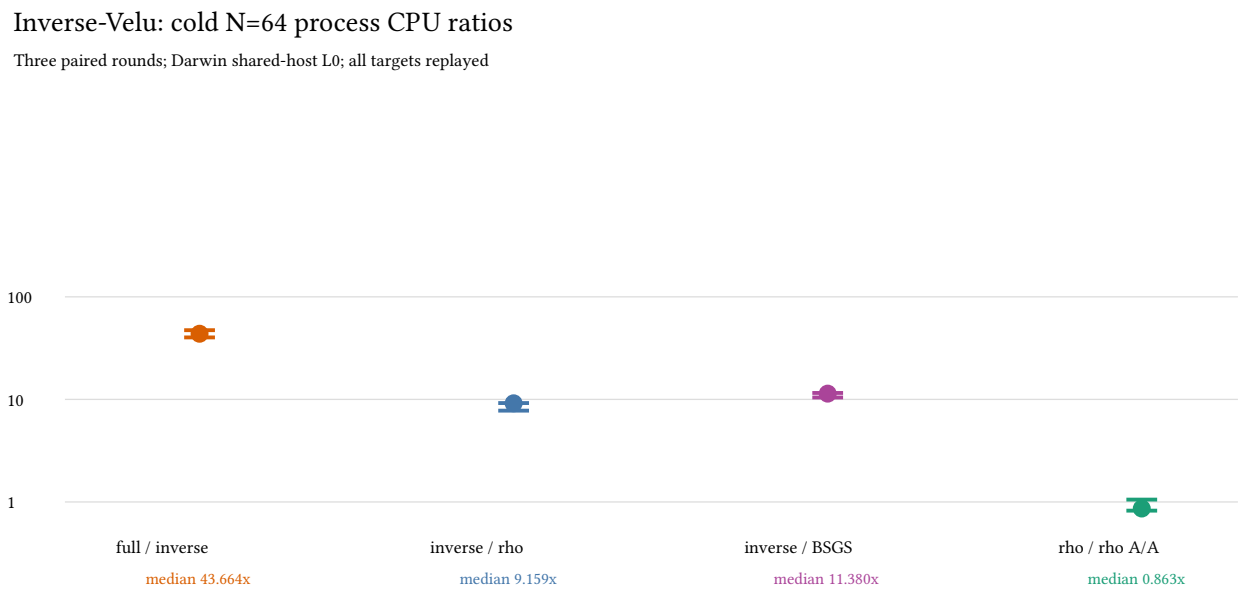
Public `p=1237`, `r=1201` subgroup · 2026-10-16 · native Rust.

The inverse x-coordinate cubic selects source witnesses for a different-`j` degree-three Vélu destination. The oracle scanned 29 destination points, made 29 polynomial calls and 22 split attempts, checked 28 complete maps, and selected the same eight factor-base columns as a 1,531,275-point source census.



Five frozen arms ran at 1/4/16/64 public targets over three paired rounds. All 60 children exited zero. An independent Rust checker replayed 1,275 scalar outputs and 6,126 relation witnesses. Across all 255 paired inverse/full target positions, support keys, trials, rank, pair entries and recovered scalars matched. The 368-file raw manifest passed SHA-256 verification. The largest inverse-child RSS at 64 targets was 9,125,888 bytes under the 512 MiB cap.

At 64 targets, counted group-cost lower bounds per target were `S_group=151.595` inverse, `1532.302` full census, `4.071` signed rho and `1.315` BSGS. Paired full/inverse was 10.108, while inverse/rho was 37.235. Complete portable `S` remains unset because native work outside the counted group axis is not fully priced.



The graph reports paired whole-child process CPU: full/inverse 43.664, inverse/rho 9.159, inverse/BSGS 11.380, and rho A/A 0.863 at 64 targets. The host was shared L0; its CPU-model `sysctl` was denied, and the host receipt preserves that limit. No L2 wall result is inferred. `REPORT.md` links the frozen spec, raw outputs, replay, analysis, source, identity proof and next-test gates.